

Binding Number, Cycles and Cliques

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Abstract

I discuss the binding number of a graph and Woodall's conjecture that binding number at least $3/2$ implies a graph is pancyclic.

My favorite conjecture was (or rather is) one about binding number. In this note I discuss it and related questions about that parameter. This area has special meaning for me, since I worked with both my advisors on these questions. Even though the original conjecture is solved, there is more here that I would still love to solve, or at least, see solved.

1 The Binding Number

The binding number of a graph was introduced in 1973 by Woodall in a seminal paper [21]. We use $N(S)$ to denote the open neighborhood of S (the set of all vertices in the graph that have a neighbor in S). Then Woodall defined:

$$\text{bind}(G) = \min \left\{ \frac{|N(S)|}{|S|} : \emptyset \subsetneq S \subseteq V(G), N(S) \neq V(G) \right\}.$$

Two simple upper bounds are pervasive. The first is obtained by choosing $S = V(G) - N(v)$ where v is a vertex of minimum degree. The second is obtained by choosing S to be a maximum independent set.

Observation 1 *For any graph G of order n , minimum degree δ , and independence number α ,*

- (a) $\text{bind}(G) \leq (n-1)/(n-\delta)$,
- (b) $\text{bind}(G) \leq (n-\alpha)/\alpha$.

For example, equality in part (a) is achieved by paths of even order, cycles of odd order, and δ -connected claw-free graphs provided $\delta \geq 3$ and $\delta \neq n-2$ [8]. Also, for any fixed p , the Erdős-Rényi random graph $G(n, p)$ almost surely has equality in part (a) [22]. Equality in part (b) is achieved for cycles of even order and for complete graphs. (Note that if one allowed $N(S) = V(G)$ in the definition of binding number, then one could take $S = V(G)$ and the parameter would always be at most 1—and in fact exactly 1 for many graphs.)

There are also some related lower bounds:

Observation 2 *For any graph G of order n , minimum degree δ and maximum degree Δ ,*
(a) $\text{bind}(G) \geq \delta/(n - \delta)$,
(b) $\text{bind}(G) \geq \delta/\Delta$.

For example, there is equality in both bounds for stars. Importantly, part (b) implies that any regular graph has binding number at least 1. Indeed it follows that any graph with a k -factor for any k has binding number at least 1.

Another important result is that Cunningham [7] showed how to compute the binding number in polynomial time.

2 The Original Conjecture

Woodall's wonderful (and lengthy!) result was that binding number at least $3/2$ forces hamiltonicity:

Theorem 1 [21] *Any graph with binding number at least $3/2$ is hamiltonian.*

The bound is best possible due to the graphs given by the join $(r - 1)K_1 \vee rK_2$. These graphs are nonhamiltonian (they are not even 1-tough). Further, their binding number is $(3r - 2)/(2r - 1)$ (achieving equality in Observation 1a), which tends to $3/2$ from below as $r \rightarrow \infty$.

His original paper [21] also introduced the Hopping Lemma. That lemma has proven useful in other cycle results; for example, it provided the key idea in the proof of Theorem 5 below to show the existence of a cycle of length $n - 1$.

At the same time, Woodall conjectured that the same threshold implies pancyclicity:

Conjecture 1 [21] *Any graph with binding number at least $3/2$*
(a) has a triangle; and indeed
(b) is pancyclic.

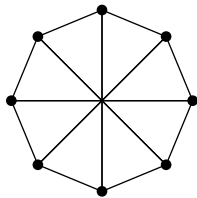
A decade or so later, the conjecture was proved by Shi Ronghua:

Theorem 2 [18] *Any graph with binding number at least $3/2$ has a triangle.*

Theorem 3 [19] *Any graph with binding number at least $3/2$ is pancyclic.*

Theorem 2 is best possible because of a family of triangle-free graphs H_k . Let k be a positive integer. Then H_k has $3k - 1$ vertices and is k -regular. These graphs have appeared in several guises. One description is that H_k is the complement of the power of the cycle

C_{3k-1}^{k-1} (see [21]). Another is that H_k is the circulant $C_{3k-1}[1, 4, \dots]$ (see e.g. [6]). For example, H_1 is K_2 , H_2 is C_5 , and H_3 is the four-rung Mobius ladder shown below. It can be shown that they achieve equality in Observation 1a; that is, their binding number is $(3k - 2)/(2k - 1)$.



The papers [18, 19] did not show up in a literature search I did. They are still not listed in Math Reviews. In fact, I learned about them in a strange way. For, I was in Kalamazoo in 1992, about to talk about my partial results on Conjecture 1, when I got an email. It was from a person who had attended the seminar in which I had given a practice talk a few weeks before. The email was along the lines of:

I was looking at this journal in the library, and I noticed the next paper to the one I was reading was about the problem you are working on. . .

(Of course, who reads bound journal copies anymore?) Anyway, I learned not to check my email just before a talk. . .

3 More About Triangles

It should be noted that the upper bounds of Observation 1 immediately imply that if the binding number is at least $3/2$, then (a) the minimum degree is at least $(n + 2)/3$, and (b) the graph is not bipartite (since its independence number is at most $2n/5$).

In response to the same seminar talk, Kleitman provided a beautiful and short proof for Theorem 2. The key idea was to establish a structural result about triangle-free graphs with high minimum degree: if they have minimum degree at least $(n + 2)/3$, they are homomorphic to a smaller triangle-free graph. Indeed, they are spanning subgraphs of maximal triangle-free graphs that have two nonadjacent vertices with the same neighborhood:

Theorem 4 [9] *A maximal triangle-free graph of order n and minimum degree $\delta \geq (n + 2)/3$ has a set I of independent vertices with identical neighborhoods where $|I| \geq 3\delta - n$.*

Here is the proof from the paper.

PROOF. Define two vertices as similar if they have the same neighborhood, and for vertex x define $s(x)$ as the number of vertices similar to x (itself included). If every two nonadjacent vertices are similar, then G is a complete multipartite graph. Indeed G is a complete bipartite graph, and the conclusion of the theorem holds for any vertex v of minimum degree.

Otherwise there exist vertices a and b that are nonadjacent and dissimilar. Let a and b be such a pair for which the overlap $|N(a) \cap N(b)|$ is maximized. Since a and b are dissimilar, there is a vertex x in $N(a) - N(b)$ say. Observe that $N(x) \cap N(b)$ is nonempty; otherwise the edge xb may be added to G without producing a triangle.

There are two cases:

(1) *There are vertices y_1 and y_2 in $N(x) \cap N(b)$ such that y_1 and y_2 are dissimilar.* Since G is triangle-free, the two sets $N(x) \cup N(b)$ and $N(y_1) \cup N(y_2)$ are disjoint. Likewise, the two sets $N(a) \cap N(b)$ and $N(x) \cap N(b)$ are disjoint. By our choice of the pair $\{a, b\}$, it holds that $|N(y_1) \cap N(y_2)| \leq |N(a) \cap N(b)|$. Hence

$$\begin{aligned} n &\geq |N(x) \cup N(b)| + |N(y_1) \cup N(y_2)| \\ &= \deg(x) + \deg(b) - |N(x) \cap N(b)| + \deg(y_1) + \deg(y_2) - |N(y_1) \cap N(y_2)| \\ &\geq \deg(x) + \deg(y_1) + \deg(y_2) + (\deg(b) - |N(x) \cap N(b)| - |N(a) \cap N(b)|) \\ &\geq \deg(x) + \deg(y_1) + \deg(y_2). \end{aligned}$$

Thus $\delta \leq n/3$, a contradiction.

(2) *All the vertices in $N(x) \cap N(b)$ are similar.* Let y be a vertex in the set $Y = N(x) \cap N(b)$. Note that $s(y) = |Y|$, and that $y \notin N(a)$. We may assume that the vertices in $X = N(y) \cap N(a)$ are similar, otherwise we are back in Case 1. Note that $x \in X$.

Since G is triangle-free, the two sets $N(x) \cup N(b)$ and $N(y)$ are disjoint. Thus

$$n \geq \deg(x) + \deg(b) - |Y| + \deg(y).$$

Similarly, $n \geq \deg(y) + \deg(a) - |X| + \deg(x)$. Addition of these two inequalities yields:

$$s(x) + s(y) + 2n \geq 2\deg(x) + 2\deg(y) + 2\delta.$$

Thus the statement of the theorem holds either for $v = x$ or for $v = y$. QED

The theorem is best possible; for example the graphs H_k from earlier show that to guarantee two similar vertices one needs $\delta \geq (n + 2)/3$. A few lines of algebra yields Theorem 2; see [9].

I have no idea what the “equivalent” result for K_4 -free graphs is, or how to establish it.

4 More About Pancyclic Graphs

A year later I talked about some more partial results (obtaining a large number of cycle lengths) at a conference in Memphis. There Stephan Brandt spoke about his work with Ralph Faudree of a similar flavor. We met. When combined and refined, these results established that actually, the pancyclic conjecture is very true: the full power of the binding number is only needed to establish the end points of the spectrum.

Theorem 5 [5, 3] *Any graph with minimum degree at least $(n+2)/3$, a triangle and a hamilton cycle, is pancyclic.*

This direction was motivated by an earlier paper by Amar et al. [1] who proved a conjecture of Häggkvist [11] that the above theorem holds for minimum degree more than $2n/5$. In fact, in [5] we showed that the conclusion holds for minimum degree at least $n/4 + O(1)$.

Which takes me to another favorite conjecture: a bipartite version of this. It is hardly surprising that one can prove, for some small c , that a hamilton bipartite graph with minimum degree at least $cn + O(1)$ has cycles of all even lengths, sometimes called bipancyclic. However, Schmeichel and Mitchem [17] asked whether for every $c > 0$ that every sufficiently large bipartite hamiltonian graph of minimum degree cn is bipancyclic. Note that linear minimum degree in bipartite graphs automatically gives small cycles. (That is, for any $\varepsilon > 0$ and any positive even integer k , every sufficiently large bipartite graph with minimum degree at least εn contains all even cycles up to length k .) Proving the existence of all the large cycle lengths, on the other hand, seems much harder. It is conjectured that one can go even further; namely, that there is some sublinear minimum degree that guarantees bipancyclicity, and indeed, perhaps even the following is true (I am not sure if this particular formulation appears in the literature):

Conjecture 2 *Any hamiltonian bipartite graph of order n with minimum degree $\delta \gg \sqrt{n}$ has cycles of all even lengths.*

There are partial results that give a large number of cycles, but not all; see for example [10, 20]. One should also note that sublinear minimum degree does not work for general graphs. That is, the $1/3$ of Theorem 5 cannot be reduced arbitrarily (see the discussion in [5]). In a different direction, Hu et al. [12] defined a bipartite version of binding number and determined the optimum threshold for a balanced bipartite graph to be bipancyclic.

5 And Cliques Too

Anyway, back to binding number. For quite a while, I thought that maybe binding number at least 2 implies the graph contains a K_4 . But, actually it is not that hard to come up with a counterexample.

Brandt [4] (and later Lyle [16]) observed that one can create such a graph by the join of graphs H and I , where H is triangle-free and I is an independent set. There is an obvious candidate for such graphs H : the extremal graphs H_k that showed that Theorem 2 is sharp. It is simple algebra to optimize the size of the independent set to show that:

Theorem 6 [4, 16] *There is a family of K_4 -free graphs with binding number approaching $(3 + \sqrt{33})/4$ from below.*

It is unresolved whether this is then best possible. (I hesitate to make another easily refuted conjecture...)

However, Brandt has conjectured that for r sufficiently large, all K_r -free graphs have binding number at most $r - 2$.

Conjecture 3 [4] *For all r sufficiently large, any graph with binding number more than $r - 2$ contains a K_r .*

The threshold $r - 2$ cannot be lower because of several constructions. One example is the tensor/direct product $K_{r-1} \times K_m$ for $m \geq r^2 - 4 + 5$. (For this range, that product has binding number equal to the minimum degree bound of Observation 1; see [14].)

Indeed, it might even be the case that $r \geq 5$ suffices. For example, attempts to add two independent sets to some H_k , or to take the join of two H_k , do of course produce a K_5 -free graph, but it does not have binding number over 3. Kane et al. [13] observed that $\text{bind}(G) > r - 4/3$ guarantees a K_r . Improvements were obtained by Brandt [4] and by us in [16], but the conjecture remains open

In general it is not known what binding number is needed for certain subgraphs. However, a lot has been written about factors and their generalization, such as optimal bounds for k -factors given by Katerinis and Woodall [15]. Indeed, the original motivation for Woodall was a result by Anderson on matchings. Translated into the terminology of the parameter that Woodall introduced, Anderson [2] showed:

Theorem 7 [2] *Any graph of even order and binding number at least $4/3$ has a perfect matching.*

6 The Future of Binding Number

There are now several hundred papers written on binding number. The main areas of study, apart from thresholds for the existence of various subgraphs discussed above, have included extremal results and computations for classes of graphs such as products. In addition, it has been proposed and studied as a “measure of vulnerability” of a network. Hopefully, results on the binding number will continue to bloom.

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